Supplementary Information

We provide additional experimental results, sensitivity analyses and details of the NH3/H2 measurement apparatus and acquisition protocol. Numerical summaries use ten predefined seeds and are conditional on the observed test exposures. The accompanying data exports retain unrounded results.

Table S1. Fold-level MAE (ppm), mean ± seed SD

Gas	Fold	Pairs	Policy	MAE
CO	1	21	Fine-tuned	70.88 ± 17.27
Ethylene	1	21	Fine-tuned	2.456 ± 0.570
CO	1	21	Frozen	81.39 ± 14.49
Ethylene	1	21	Frozen	2.914 ± 0.780
CO	1	21	Scratch	83.56 ± 24.03
Ethylene	1	21	Scratch	3.159 ± 1.091
CO	2	18	Fine-tuned	76.25 ± 15.60
Ethylene	2	18	Fine-tuned	1.675 ± 0.184
CO	2	18	Frozen	74.09 ± 16.26
Ethylene	2	18	Frozen	1.517 ± 0.224
CO	2	18	Scratch	105.82 ± 18.97
Ethylene	2	18	Scratch	1.989 ± 0.283
CO	3	22	Fine-tuned	48.52 ± 9.40
Ethylene	3	22	Fine-tuned	1.686 ± 0.328
CO	3	22	Frozen	47.18 ± 12.14
Ethylene	3	22	Frozen	1.711 ± 0.357
CO	3	22	Scratch	80.47 ± 31.42
Ethylene	3	22	Scratch	2.684 ± 0.590
H2	1	57	Fine-tuned	45.39 ± 4.30
NH3	1	57	Fine-tuned	41.99 ± 5.37
H2	1	57	Frozen	39.17 ± 3.27
NH3	1	57	Frozen	37.38 ± 3.48
H2	1	57	Scratch	70.00 ± 15.43
NH3	1	57	Scratch	52.04 ± 9.25
H2	2	57	Fine-tuned	37.28 ± 7.90
NH3	2	57	Fine-tuned	26.53 ± 5.04
H2	2	57	Frozen	32.11 ± 4.01
NH3	2	57	Frozen	25.24 ± 4.21
H2	2	57	Scratch	44.98 ± 11.13
NH3	2	57	Scratch	30.59 ± 8.63
H2	3	57	Fine-tuned	35.13 ± 2.95
NH3	3	57	Fine-tuned	17.05 ± 4.34
H2	3	57	Frozen	37.40 ± 4.31
NH3	3	57	Frozen	17.72 ± 2.31
H2	3	57	Scratch	46.91 ± 7.11
NH3	3	57	Scratch	20.94 ± 4.31
H2	4	57	Fine-tuned	26.36 ± 4.22
NH3	4	57	Fine-tuned	18.83 ± 3.31

Gas	Fold	Pairs	Policy	MAE
H2	4	57	Frozen	30.66 ± 4.48
NH3	4	57	Frozen	22.77 ± 4.16
H2	4	57	Scratch	25.70 ± 3.99
NH3	4	57	Scratch	16.22 ± 3.48
H2	5	57	Fine-tuned	55.99 ± 4.86
NH3	5	57	Fine-tuned	38.39 ± 2.50
H2	5	57	Frozen	59.73 ± 4.79
NH3	5	57	Frozen	44.10 ± 3.37
H2	5	57	Scratch	66.43 ± 8.92
NH3	5	57	Scratch	44.66 ± 5.92
H2	6	57	Fine-tuned	31.60 ± 2.29
NH3	6	57	Fine-tuned	17.36 ± 1.54
H2	6	57	Frozen	34.77 ± 2.70
NH3	6	57	Frozen	19.26 ± 1.76
H2	6	57	Scratch	41.20 ± 4.27
NH3	6	57	Scratch	21.59 ± 3.25

Table S2. Seed-count sensitivity within the revised primary protocol

Gas	First 3 gain %	All 10 gain %	Later 7 gain %	Positive / 10
CO	28.48	27.50	27.06	10
Ethylene	26.09	25.73	25.57	9
H2	18.40	21.49	22.72	10
NH3	13.30	13.91	14.15	9

Table S2b. Secondary full-budget fixed split and drift seed sensitivity

Analysis	Gas	First 3 gain %	All 10 gain %	Later 7 gain %
fixed_split	CO	27.23	25.43	24.71
fixed_split	Ethylene	31.14	24.31	21.01
fixed_split	H2	27.22	23.29	21.52
fixed_split	NH3	32.44	19.59	12.96
drift_partition_8	MacroMAE	25.61	18.80	15.63
drift_partition_9	MacroMAE	17.97	18.35	18.52
drift_partition_10	MacroMAE	-11.95	-0.50	4.17

Prefixes use the predefined order 42, 142, …, 942; all ten remain the reporting set. A random-order 95% envelope is a subset diagnostic, not a confidence interval. More seeds change an estimated mean; they do not weaken a trained model or increase the number of independent physical exposures.

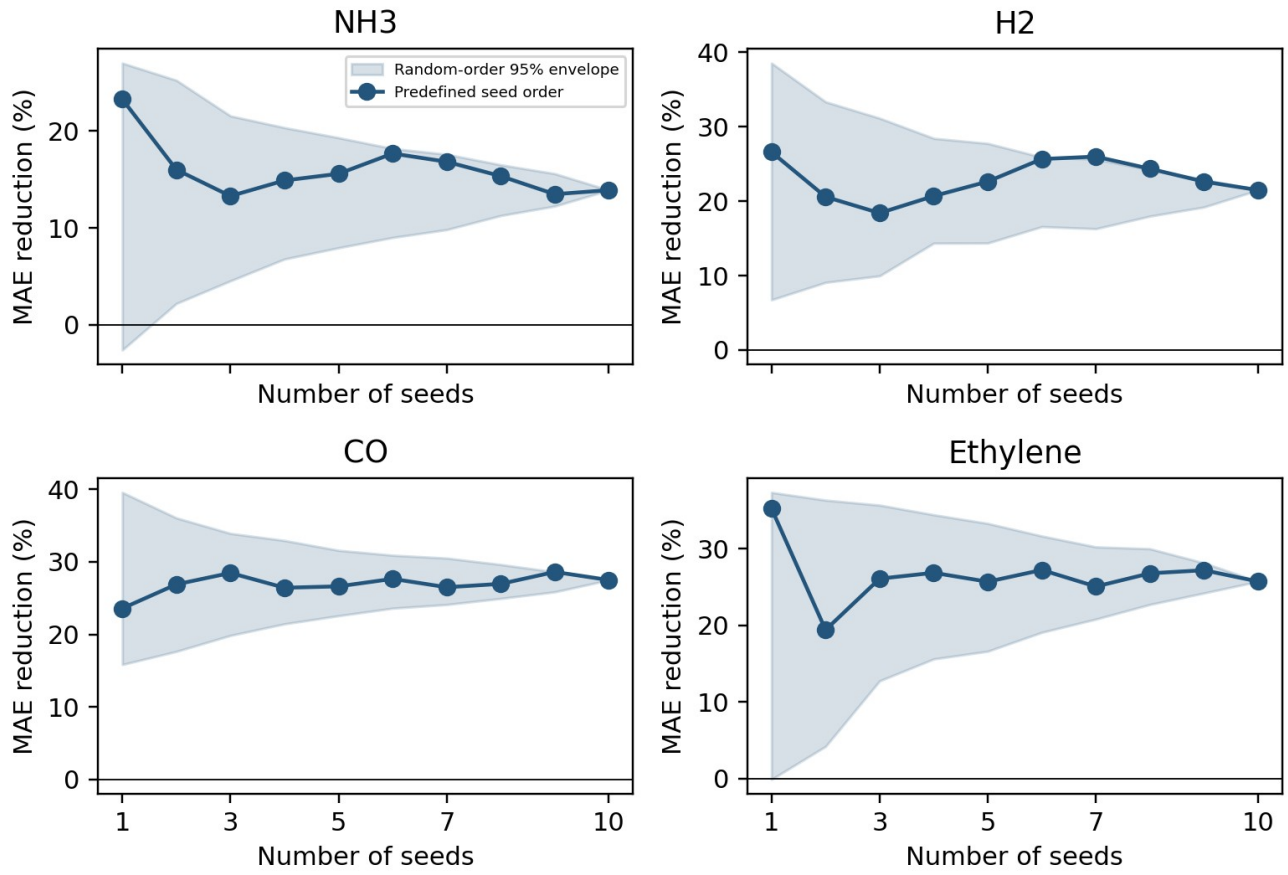


Fig. S1. Primary relative MAE reduction against scratch as seeds accumulate; shading shows 2.5th–97.5th percentiles of 2,000 random subsets at each count.

Table S2a. Every primary seed prefix (errors and intervals in ppm)

Gas	Seeds	Scratch	Fine-tuned	Gain %	Difference 95% interval
CO	1	94.72	72.38	23.58	[-10.73, 40.80]
Ethylene	1	2.754	1.783	35.24	[0.536, 1.689]
H2	1	54.39	39.92	26.61	[2.49, 27.40]
NH3	1	31.34	24.04	23.28	[1.27, 11.46]
CO	2	97.23	71.07	26.90	[2.83, 39.51]
Ethylene	2	2.495	2.011	19.41	[-0.222, 1.355]
H2	2	48.85	38.79	20.58	[1.01, 22.83]
NH3	2	30.62	25.71	16.02	[-2.62, 10.99]
CO	3	93.44	66.83	28.48	[3.80, 41.93]
Ethylene	3	2.693	1.990	26.09	[-0.002, 1.502]
H2	3	46.74	38.14	18.40	[2.19, 19.06]
NH3	3	29.61	25.67	13.30	[-1.07, 9.43]
CO	4	90.43	66.53	26.43	[4.89, 40.50]
Ethylene	4	2.760	2.018	26.86	[0.156, 1.397]
H2	4	49.09	38.93	20.69	[3.10, 20.90]
NH3	4	30.86	26.26	14.91	[-0.50, 9.69]
CO	5	87.55	64.24	26.62	[6.63, 37.84]

Gas	Seeds	Scratch	Fine-tuned	Gain %	Difference 95% interval
Ethylene	5	2.670	1.984	25.70	[0.156, 1.265]
H2	5	49.39	38.20	22.64	[4.36, 20.71]
NH3	5	30.92	26.10	15.59	[0.32, 9.26]
CO	6	89.41	64.68	27.66	[8.76, 38.90]
Ethylene	6	2.703	1.967	27.23	[0.259, 1.230]
H2	6	50.83	37.78	25.67	[5.55, 23.25]
NH3	6	31.54	25.97	17.68	[1.43, 10.28]
CO	7	86.97	63.92	26.50	[7.37, 36.65]
Ethylene	7	2.622	1.965	25.07	[0.129, 1.168]
H2	7	51.31	37.98	25.99	[6.14, 22.22]
NH3	7	31.77	26.42	16.84	[1.56, 9.55]
CO	8	89.21	65.15	26.97	[10.41, 36.09]
Ethylene	8	2.670	1.954	26.79	[0.226, 1.243]
H2	8	50.54	38.23	24.36	[4.64, 21.09]
NH3	8	31.15	26.36	15.36	[0.80, 8.97]
CO	9	91.21	65.11	28.62	[12.06, 39.43]
Ethylene	9	2.670	1.944	27.20	[0.260, 1.201]
H2	9	49.80	38.53	22.63	[3.73, 19.95]
NH3	9	30.91	26.74	13.49	[0.01, 8.48]
CO	10	89.95	65.22	27.50	[12.85, 37.53]
Ethylene	10	2.611	1.939	25.73	[0.247, 1.143]
H2	10	49.20	38.63	21.49	[3.57, 18.78]
NH3	10	31.01	26.69	13.91	[0.38, 8.30]

Table S3. Random minus farthest-point design MAE (ppm)

Gas	Pairs	Policy	Random	Farthest	Difference [95% interval]
CO	5	Scratch	179.22	163.06	16.16 [-31.10, 53.12]
Ethylene	5	Scratch	4.890	5.226	-0.335 [-1.621, 0.930]
CO	10	Scratch	144.05	156.04	-11.99 [-46.20, 25.05]
Ethylene	10	Scratch	3.881	3.610	0.271 [-0.951, 1.591]
CO	20	Scratch	83.83	89.30	-5.47 [-26.51, 14.46]
Ethylene	20	Scratch	2.489	2.465	0.024 [-0.422, 0.504]
CO	24	Scratch	63.63	63.63	0.00 [0.00, 0.00]
Ethylene	24	Scratch	2.149	2.149	0.000 [0.000, 0.000]
H2	5	Scratch	159.51	161.51	-2.00 [-19.61, 11.72]
NH3	5	Scratch	98.51	86.39	12.12 [-13.67, 38.24]
H2	10	Scratch	137.23	124.16	13.07 [-14.68, 36.59]
NH3	10	Scratch	69.60	67.52	2.08 [-23.07, 23.84]
H2	20	Scratch	70.48	70.73	-0.25 [-24.56, 22.33]
NH3	20	Scratch	42.35	36.84	5.51 [-2.55, 13.76]
H2	40	Scratch	43.21	51.53	-8.32 [-11.53, -4.60]
NH3	40	Scratch	25.40	29.86	-4.46 [-6.35, -1.66]
H2	57	Scratch	41.20	41.20	0.00 [0.00, 0.00]
NH3	57	Scratch	21.59	21.59	0.00 [0.00, 0.00]
CO	5	Fine-tuned	143.07	111.46	31.61 [13.34, 51.92]
Ethylene	5	Fine-tuned	4.323	3.468	0.855 [-0.595, 2.151]

Gas	Pairs	Policy	Random	Farthest	Difference [95% interval]
CO	10	Fine-tuned	108.99	80.99	28.00 [-4.59, 62.23]
Ethylene	10	Fine-tuned	3.081	2.418	0.663 [0.069, 1.392]
CO	20	Fine-tuned	54.22	47.42	6.80 [0.99, 12.18]
Ethylene	20	Fine-tuned	1.987	1.843	0.144 [-0.137, 0.457]
CO	24	Fine-tuned	47.45	47.45	0.00 [0.00, 0.00]
Ethylene	24	Fine-tuned	1.627	1.627	0.000 [0.000, 0.000]
H2	5	Fine-tuned	143.53	128.91	14.62 [-5.72, 35.31]
NH3	5	Fine-tuned	89.75	86.02	3.74 [-17.95, 28.62]
H2	10	Fine-tuned	103.24	81.88	21.36 [-0.61, 42.36]
NH3	10	Fine-tuned	52.62	38.92	13.69 [5.83, 20.77]
H2	20	Fine-tuned	47.14	48.38	-1.24 [-9.01, 5.32]
NH3	20	Fine-tuned	34.40	29.53	4.87 [1.89, 8.05]
H2	40	Fine-tuned	36.78	41.67	-4.89 [-7.31, -2.45]
NH3	40	Fine-tuned	21.85	25.22	-3.37 [-5.50, -1.12]
H2	57	Fine-tuned	31.60	31.60	0.00 [0.00, 0.00]
NH3	57	Fine-tuned	17.36	17.36	0.00 [0.00, 0.00]

The export column scratch_mae in this reused paired-comparison routine is the comparator mean (random design here); transfer_mae is the farthest-point mean. Positive differences favor farthest-point design. These exploratory comparisons are not multiplicity-adjusted.

Table S4. Generic SSL minus sensor-aware MAE (ppm)

Gas	Pairs	Generic	Sensor-aware	Difference [95% interval]
CO	5	119.26	111.46	7.80 [-6.70, 21.72]
Ethylene	5	3.996	3.468	0.529 [-0.592, 1.703]
CO	10	93.05	80.99	12.05 [-9.66, 30.80]
Ethylene	10	2.735	2.418	0.317 [-0.119, 0.694]
CO	20	51.08	47.42	3.66 [-0.87, 7.73]
Ethylene	20	1.622	1.843	-0.221 [-0.446, 0.015]
CO	24	52.52	47.45	5.08 [-2.31, 12.72]
Ethylene	24	1.437	1.627	-0.189 [-0.336, -0.034]
H2	5	121.62	128.91	-7.30 [-30.60, 12.11]
NH3	5	74.57	86.02	-11.45 [-31.77, 9.40]
H2	10	91.13	81.88	9.25 [-0.74, 19.71]
NH3	10	39.49	38.92	0.56 [-3.19, 4.20]
H2	20	49.92	48.38	1.54 [-2.56, 5.51]
NH3	20	31.61	29.53	2.08 [0.38, 3.91]
H2	40	41.33	41.67	-0.34 [-2.49, 2.11]
NH3	40	24.17	25.22	-1.05 [-2.19, 0.11]
H2	57	35.18	31.60	3.58 [1.53, 5.40]
NH3	57	16.13	17.36	-1.23 [-2.46, -0.16]

The control is a compact reimplementation inspired by PatchTST, not the official forecasting configuration. See Section IV for architectural differences. PLSR candidates use 1, 2, 3, 5, 8 or 12 components when below the fitting-sample/feature bound. RBF SVR searches C ∈ {0.1, 1, 10}, gamma ∈ {scale, 0.001}, epsilon ∈ {0.01, 0.1}, on fixed-normalized targets. Selection minimizes adaptation-validation normalized MAE. Extra Trees uses 300 trees and minimum leaf size 2. XGBoost uses 300 trees, depth 3, learning rate 0.03, and row/feature fractions 0.9. Test predictions are clipped to the same target ranges as neural models.

Table S4a. All-budget baseline MAE (ppm), mean ± SD

Gas	Pairs	Model	MAE
CO	5	Extra Trees	132.28 ± 25.01
Ethylene	5	Extra Trees	3.318 ± 0.578
CO	5	Fine-tuned	111.46 ± 23.96
Ethylene	5	Fine-tuned	3.468 ± 1.248
CO	5	Generic SSL	119.26 ± 23.91
Ethylene	5	Generic SSL	3.996 ± 1.136
CO	5	PLSR	143.45 ± 41.09
Ethylene	5	PLSR	3.427 ± 0.858
CO	5	Scratch	163.06 ± 47.96
Ethylene	5	Scratch	5.226 ± 1.469
CO	5	SVR	131.59 ± 5.17
Ethylene	5	SVR	3.812 ± 0.378
CO	5	XGBoost	164.46 ± 41.00
Ethylene	5	XGBoost	4.956 ± 1.288
CO	10	Extra Trees	121.83 ± 26.67
Ethylene	10	Extra Trees	2.438 ± 0.303
CO	10	Fine-tuned	80.99 ± 27.98
Ethylene	10	Fine-tuned	2.418 ± 0.484
CO	10	Generic SSL	93.05 ± 28.68
Ethylene	10	Generic SSL	2.735 ± 0.560
CO	10	PLSR	76.36 ± 49.19
Ethylene	10	PLSR	2.703 ± 0.579
CO	10	Scratch	156.04 ± 27.60
Ethylene	10	Scratch	3.610 ± 1.421
CO	10	SVR	102.30 ± 8.29
Ethylene	10	SVR	3.032 ± 0.380
CO	10	XGBoost	99.25 ± 35.41
Ethylene	10	XGBoost	4.701 ± 1.142
CO	20	Extra Trees	104.17 ± 10.63
Ethylene	20	Extra Trees	1.616 ± 0.266
CO	20	Fine-tuned	47.42 ± 7.79
Ethylene	20	Fine-tuned	1.843 ± 0.506
CO	20	Generic SSL	51.08 ± 6.44
Ethylene	20	Generic SSL	1.622 ± 0.304
CO	20	PLSR	31.57 ± 8.90
Ethylene	20	PLSR	2.056 ± 0.424
CO	20	Scratch	89.30 ± 39.24
Ethylene	20	Scratch	2.465 ± 0.745
CO	20	SVR	51.12 ± 10.66
Ethylene	20	SVR	2.212 ± 0.213
CO	20	XGBoost	64.95 ± 9.22
Ethylene	20	XGBoost	2.903 ± 0.914
CO	24	Extra Trees	90.46 ± 7.17
Ethylene	24	Extra Trees	1.398 ± 0.197
CO	24	Fine-tuned	47.45 ± 7.38
Ethylene	24	Fine-tuned	1.627 ± 0.275

Gas	Pairs	Model	MAE
CO	24	Generic SSL	52.52 ± 12.59
Ethylene	24	Generic SSL	1.437 ± 0.275
CO	24	PLSR	25.57 ± 8.48
Ethylene	24	PLSR	1.994 ± 0.377
CO	24	Scratch	63.63 ± 10.68
Ethylene	24	Scratch	2.149 ± 0.359
CO	24	SVR	53.01 ± 7.26
Ethylene	24	SVR	2.013 ± 0.226
CO	24	XGBoost	55.90 ± 11.13
Ethylene	24	XGBoost	2.491 ± 0.352
H2	5	Extra Trees	147.77 ± 35.20
NH3	5	Extra Trees	56.11 ± 9.00
H2	5	Fine-tuned	128.91 ± 16.84
NH3	5	Fine-tuned	86.02 ± 34.16
H2	5	Generic SSL	121.62 ± 35.14
NH3	5	Generic SSL	74.57 ± 31.27
H2	5	PLSR	156.59 ± 14.73
NH3	5	PLSR	51.90 ± 10.24
H2	5	Scratch	161.51 ± 22.40
NH3	5	Scratch	86.39 ± 36.11
H2	5	SVR	147.89 ± 36.56
NH3	5	SVR	67.15 ± 11.23
H2	5	XGBoost	163.11 ± 27.20
NH3	5	XGBoost	98.92 ± 24.61
H2	10	Extra Trees	108.23 ± 26.56
NH3	10	Extra Trees	48.09 ± 4.98
H2	10	Fine-tuned	81.88 ± 19.54
NH3	10	Fine-tuned	38.92 ± 4.56
H2	10	Generic SSL	91.13 ± 30.27
NH3	10	Generic SSL	39.49 ± 5.12
H2	10	PLSR	133.49 ± 29.68
NH3	10	PLSR	44.47 ± 6.31
H2	10	Scratch	124.16 ± 28.13
NH3	10	Scratch	67.52 ± 40.50
H2	10	SVR	116.17 ± 23.18
NH3	10	SVR	48.21 ± 6.82
H2	10	XGBoost	110.10 ± 13.23
NH3	10	XGBoost	69.34 ± 10.88
H2	20	Extra Trees	69.90 ± 11.10
NH3	20	Extra Trees	39.65 ± 2.44
H2	20	Fine-tuned	48.38 ± 9.70
NH3	20	Fine-tuned	29.53 ± 2.38
H2	20	Generic SSL	49.92 ± 7.00
NH3	20	Generic SSL	31.61 ± 3.35
H2	20	PLSR	116.94 ± 25.77
NH3	20	PLSR	39.09 ± 4.60
H2	20	Scratch	70.73 ± 30.45

Gas	Pairs	Model	MAE
NH3	20	Scratch	36.84 ± 7.89
H2	20	SVR	76.97 ± 20.72
NH3	20	SVR	35.54 ± 6.09
H2	20	XGBoost	83.40 ± 9.19
NH3	20	XGBoost	46.36 ± 8.16
H2	40	Extra Trees	51.73 ± 4.98
NH3	40	Extra Trees	34.14 ± 3.52
H2	40	Fine-tuned	41.67 ± 2.75
NH3	40	Fine-tuned	25.22 ± 2.10
H2	40	Generic SSL	41.33 ± 3.36
NH3	40	Generic SSL	24.17 ± 1.74
H2	40	PLSR	90.80 ± 10.20
NH3	40	PLSR	32.04 ± 5.77
H2	40	Scratch	51.53 ± 6.60
NH3	40	Scratch	29.86 ± 2.74
H2	40	SVR	38.34 ± 4.15
NH3	40	SVR	25.34 ± 3.44
H2	40	XGBoost	62.60 ± 4.85
NH3	40	XGBoost	39.78 ± 7.64
H2	57	Extra Trees	37.33 ± 2.04
NH3	57	Extra Trees	21.63 ± 1.73
H2	57	Fine-tuned	31.60 ± 2.29
NH3	57	Fine-tuned	17.36 ± 1.54
H2	57	Generic SSL	35.18 ± 2.74
NH3	57	Generic SSL	16.13 ± 1.23
H2	57	PLSR	75.50 ± 18.25
NH3	57	PLSR	28.44 ± 4.76
H2	57	Scratch	41.20 ± 4.27
NH3	57	Scratch	21.59 ± 3.25
H2	57	SVR	31.37 ± 4.53
NH3	57	SVR	20.74 ± 1.62
H2	57	XGBoost	48.65 ± 2.38
NH3	57	XGBoost	26.96 ± 4.61

Table S5. All-policy per-gas drift errors at 20 labels/gas

Partition	Gas	Policy	MAE ppm	NMAE %
8	Acetaldehyde	Constant	11.79 ± 7.58	4.06 ± 2.61
8	Acetone	Constant	57.28 ± 12.80	11.74 ± 2.62
8	Ammonia	Constant	17.06 ± 4.97	1.80 ± 0.52
8	Ethanol	Constant	77.11 ± 7.16	13.07 ± 1.21
8	Ethylene	Constant	94.67 ± 11.81	32.65 ± 4.07
8	Toluene	Constant	59.57 ± 6.19	62.71 ± 6.52
8	Acetaldehyde	Fine-tuned	16.40 ± 7.04	5.65 ± 2.43
8	Acetone	Fine-tuned	33.23 ± 1.62	6.81 ± 0.33
8	Ammonia	Fine-tuned	31.75 ± 8.84	3.34 ± 0.93
8	Ethanol	Fine-tuned	2.87 ± 3.09	0.49 ± 0.52
8	Ethylene	Fine-tuned	14.63 ± 4.06	5.04 ± 1.40

Partition	Gas	Policy	MAE ppm	NMAE %
8	Toluene	Fine-tuned	17.82 ± 7.90	18.76 ± 8.32
8	Acetaldehyde	Frozen	13.44 ± 3.77	4.63 ± 1.30
8	Acetone	Frozen	31.53 ± 1.41	6.46 ± 0.29
8	Ammonia	Frozen	30.10 ± 8.50	3.17 ± 0.89
8	Ethanol	Frozen	3.48 ± 3.28	0.59 ± 0.56
8	Ethylene	Frozen	16.43 ± 3.21	5.67 ± 1.11
8	Toluene	Frozen	19.55 ± 6.72	20.58 ± 7.08
8	Acetaldehyde	Scratch	20.21 ± 11.03	6.97 ± 3.81
8	Acetone	Scratch	32.11 ± 1.93	6.58 ± 0.39
8	Ammonia	Scratch	38.91 ± 12.96	4.10 ± 1.36
8	Ethanol	Scratch	6.22 ± 3.63	1.05 ± 0.62
8	Ethylene	Scratch	17.30 ± 5.68	5.97 ± 1.96
8	Toluene	Scratch	28.96 ± 6.90	30.49 ± 7.27
9	Acetaldehyde	Constant	10.05 ± 7.51	3.46 ± 2.59
9	Acetone	Constant	24.91 ± 13.28	5.10 ± 2.72
9	Ammonia	Constant	10.50 ± 6.96	1.11 ± 0.73
9	Ethanol	Constant	77.11 ± 7.16	13.07 ± 1.21
9	Ethylene	Constant	94.67 ± 11.81	32.65 ± 4.07
9	Toluene	Constant	59.57 ± 6.19	62.71 ± 6.52
9	Acetaldehyde	Fine-tuned	29.07 ± 8.48	10.02 ± 2.92
9	Acetone	Fine-tuned	22.19 ± 6.21	4.55 ± 1.27
9	Ammonia	Fine-tuned	16.53 ± 9.71	1.74 ± 1.02
9	Ethanol	Fine-tuned	2.39 ± 2.65	0.40 ± 0.45
9	Ethylene	Fine-tuned	12.28 ± 4.36	4.23 ± 1.50
9	Toluene	Fine-tuned	28.65 ± 11.03	30.16 ± 11.61
9	Acetaldehyde	Frozen	29.77 ± 4.49	10.27 ± 1.55
9	Acetone	Frozen	23.19 ± 5.32	4.75 ± 1.09
9	Ammonia	Frozen	17.62 ± 8.69	1.85 ± 0.92
9	Ethanol	Frozen	2.44 ± 2.87	0.41 ± 0.49
9	Ethylene	Frozen	12.71 ± 3.98	4.38 ± 1.37
9	Toluene	Frozen	27.73 ± 7.11	29.19 ± 7.48
9	Acetaldehyde	Scratch	30.22 ± 8.42	10.42 ± 2.90
9	Acetone	Scratch	22.35 ± 4.69	4.58 ± 0.96
9	Ammonia	Scratch	21.34 ± 13.22	2.25 ± 1.39
9	Ethanol	Scratch	2.00 ± 2.16	0.34 ± 0.37
9	Ethylene	Scratch	13.59 ± 5.25	4.69 ± 1.81
9	Toluene	Scratch	46.58 ± 13.62	49.03 ± 14.33
10	Acetaldehyde	Constant	76.54 ± 2.66	26.39 ± 0.92
10	Acetone	Constant	408.40 ± 8.85	83.69 ± 1.81
10	Ammonia	Constant	415.97 ± 7.50	43.79 ± 0.79
10	Ethanol	Constant	78.13 ± 2.39	13.24 ± 0.40
10	Ethylene	Constant	74.27 ± 1.49	25.61 ± 0.51
10	Toluene	Constant	33.02 ± 2.06	34.76 ± 2.17
10	Acetaldehyde	Fine-tuned	70.15 ± 6.78	24.19 ± 2.34
10	Acetone	Fine-tuned	298.21 ± 62.08	61.11 ± 12.72
10	Ammonia	Fine-tuned	341.99 ± 29.11	36.00 ± 3.06
10	Ethanol	Fine-tuned	63.48 ± 5.29	10.76 ± 0.90

Partition	Gas	Policy	MAE ppm	NMAE %
10	Ethylene	Fine-tuned	63.00 ± 10.18	21.72 ± 3.51
10	Toluene	Fine-tuned	21.48 ± 4.35	22.61 ± 4.58
10	Acetaldehyde	Frozen	72.39 ± 4.36	24.96 ± 1.50
10	Acetone	Frozen	285.49 ± 63.18	58.50 ± 12.95
10	Ammonia	Frozen	316.10 ± 48.02	33.27 ± 5.05
10	Ethanol	Frozen	65.16 ± 5.24	11.04 ± 0.89
10	Ethylene	Frozen	59.67 ± 5.23	20.57 ± 1.81
10	Toluene	Frozen	25.75 ± 6.08	27.11 ± 6.40
10	Acetaldehyde	Scratch	64.32 ± 10.02	22.18 ± 3.45
10	Acetone	Scratch	318.77 ± 21.20	65.32 ± 4.34
10	Ammonia	Scratch	341.17 ± 47.66	35.91 ± 5.02
10	Ethanol	Scratch	62.47 ± 9.20	10.59 ± 1.56
10	Ethylene	Scratch	51.07 ± 7.13	17.61 ± 2.46
10	Toluene	Scratch	16.23 ± 4.84	17.08 ± 5.09

Table S5a. Historical reporting ranges from partitions 1–6 (ppm)

Gas	Minimum	Maximum	Range
Ethanol	10.00	600.00	590.00
Ethylene	10.00	300.00	290.00
Ammonia	50.00	1000.00	950.00
Acetaldehyde	10.00	300.00	290.00
Acetone	12.00	500.00	488.00
Toluene	5.00	100.00	95.00

Table S6. Small-budget drift macro errors, mean ± SD

Partition	Labels/gas	Policy	Macro-MAE ppm	Macro-NMAE %
8	1	Constant	88.18 ± 26.42	25.28 ± 8.31
8	1	Fine-tuned	40.73 ± 24.75	16.36 ± 19.63
8	1	Frozen	45.57 ± 27.97	18.92 ± 27.04
8	1	Scratch	46.22 ± 20.34	18.12 ± 10.66
8	2	Constant	86.33 ± 28.40	29.98 ± 17.85
8	2	Fine-tuned	44.37 ± 5.14	11.02 ± 1.80
8	2	Frozen	45.97 ± 4.42	11.84 ± 2.79
8	2	Scratch	50.39 ± 5.95	15.83 ± 5.85
8	5	Constant	52.04 ± 9.09	21.60 ± 4.26
8	5	Fine-tuned	35.55 ± 7.65	9.21 ± 3.43
8	5	Frozen	34.96 ± 7.97	9.05 ± 2.69
8	5	Scratch	38.02 ± 7.28	9.84 ± 3.46
8	10	Constant	47.50 ± 5.31	17.36 ± 2.44
8	10	Fine-tuned	30.05 ± 6.09	8.37 ± 3.24
8	10	Frozen	27.18 ± 4.19	6.55 ± 0.97
8	10	Scratch	34.93 ± 6.77	9.66 ± 3.25
8	20	Constant	52.91 ± 3.07	21.00 ± 1.54
8	20	Fine-tuned	19.45 ± 3.37	6.68 ± 1.74
8	20	Frozen	19.09 ± 2.25	6.85 ± 1.34
8	20	Scratch	23.95 ± 3.46	9.19 ± 1.49
9	1	Constant	81.78 ± 26.67	24.12 ± 8.32

Partition	Labels/gas	Policy	Macro-MAE ppm	Macro-NMAE %
9	1	Fine-tuned	39.47 ± 20.58	13.56 ± 11.81
9	1	Frozen	43.25 ± 19.01	14.88 ± 14.87
9	1	Scratch	44.65 ± 20.38	17.46 ± 10.04
9	2	Constant	85.73 ± 28.18	29.90 ± 17.50
9	2	Fine-tuned	48.76 ± 5.38	11.94 ± 2.15
9	2	Frozen	50.50 ± 4.55	13.06 ± 3.40
9	2	Scratch	53.15 ± 6.03	16.61 ± 6.24
9	5	Constant	48.70 ± 9.11	20.88 ± 4.30
9	5	Fine-tuned	43.45 ± 8.11	12.78 ± 5.76
9	5	Frozen	41.33 ± 6.89	11.75 ± 3.81
9	5	Scratch	44.84 ± 4.97	11.98 ± 3.46
9	10	Constant	41.84 ± 5.14	16.17 ± 2.37
9	10	Fine-tuned	33.39 ± 7.33	9.01 ± 2.33
9	10	Frozen	30.83 ± 5.20	7.68 ± 1.13
9	10	Scratch	40.63 ± 3.54	11.28 ± 2.89
9	20	Constant	46.13 ± 3.07	19.68 ± 1.52
9	20	Fine-tuned	18.52 ± 4.60	8.52 ± 2.48
9	20	Frozen	18.91 ± 3.25	8.48 ± 1.63
9	20	Scratch	22.68 ± 2.86	11.88 ± 2.12
10	1	Constant	177.17 ± 10.32	39.52 ± 3.02
10	1	Fine-tuned	206.22 ± 24.05	47.14 ± 13.39
10	1	Frozen	209.72 ± 29.55	50.43 ± 23.17
10	1	Scratch	199.79 ± 14.71	44.84 ± 6.20
10	2	Constant	212.96 ± 15.13	51.87 ± 9.35
10	2	Fine-tuned	225.23 ± 15.18	49.28 ± 3.65
10	2	Frozen	224.86 ± 16.02	48.72 ± 4.14
10	2	Scratch	222.05 ± 15.74	48.44 ± 3.76
10	5	Constant	192.76 ± 4.52	41.03 ± 1.29
10	5	Fine-tuned	196.99 ± 23.05	42.51 ± 4.63
10	5	Frozen	207.84 ± 33.72	46.61 ± 7.37
10	5	Scratch	192.46 ± 22.93	40.09 ± 4.96
10	10	Constant	186.96 ± 2.84	38.63 ± 0.66
10	10	Fine-tuned	196.18 ± 17.97	41.83 ± 4.40
10	10	Frozen	191.79 ± 15.94	41.18 ± 3.51
10	10	Scratch	191.07 ± 18.61	40.41 ± 3.95
10	20	Constant	181.05 ± 2.13	37.91 ± 0.54
10	20	Fine-tuned	143.05 ± 12.49	29.40 ± 3.07
10	20	Frozen	137.43 ± 13.49	29.24 ± 2.67
10	20	Scratch	142.34 ± 11.06	28.12 ± 2.31

Below 20 labels/gas the predefined target is fixed-log Huber; at 20 it is historical fitted-log Huber. Therefore this primary budget series also changes scaling at 20. The fixed-log 20-label rows in Table S7 provide a like-for-like scaling control. One-label runs have no validation holdout.

Table S7. All-policy 20-label loss/scaling controls

Partition	Transform	Policy	Macro-MAE ppm	Macro-NMAE %
8	linear_huber	Fine-tuned	26.94 ± 4.16	12.49 ± 4.10
8	linear_huber	Frozen	27.67 ± 7.34	11.06 ± 3.86

Partition	Transform	Policy	Macro-MAE ppm	Macro-NMAE %
8	linear_huber	Scratch	32.50 ± 6.46	16.02 ± 6.66
8	linear_mae	Fine-tuned	22.48 ± 4.17	8.20 ± 3.21
8	linear_mae	Frozen	22.73 ± 3.55	7.55 ± 1.96
8	linear_mae	Scratch	27.01 ± 6.50	11.61 ± 4.91
8	log_fixed_huber	Fine-tuned	22.68 ± 5.59	7.65 ± 2.86
8	log_fixed_huber	Frozen	20.32 ± 4.02	6.42 ± 1.79
8	log_fixed_huber	Scratch	26.49 ± 2.40	9.07 ± 1.43
8	log_huber	Constant	52.91 ± 3.07	21.00 ± 1.54
8	log_huber	Fine-tuned	19.45 ± 3.37	6.68 ± 1.74
8	log_huber	Frozen	19.09 ± 2.25	6.85 ± 1.34
8	log_huber	Scratch	23.95 ± 3.46	9.19 ± 1.49
9	linear_huber	Fine-tuned	25.50 ± 6.16	13.09 ± 4.63
9	linear_huber	Frozen	26.49 ± 9.66	11.44 ± 4.99
9	linear_huber	Scratch	32.04 ± 7.62	17.25 ± 7.21
9	linear_mae	Fine-tuned	20.70 ± 4.75	8.91 ± 3.98
9	linear_mae	Frozen	20.28 ± 3.36	7.72 ± 2.42
9	linear_mae	Scratch	25.69 ± 6.77	12.70 ± 5.14
9	log_fixed_huber	Fine-tuned	26.11 ± 6.19	8.90 ± 3.19
9	log_fixed_huber	Frozen	23.28 ± 4.85	7.51 ± 2.02
9	log_fixed_huber	Scratch	33.68 ± 4.07	12.02 ± 2.17
9	log_huber	Constant	46.13 ± 3.07	19.68 ± 1.52
9	log_huber	Fine-tuned	18.52 ± 4.60	8.52 ± 2.48
9	log_huber	Frozen	18.91 ± 3.25	8.48 ± 1.63
9	log_huber	Scratch	22.68 ± 2.86	11.88 ± 2.12
10	linear_huber	Fine-tuned	145.56 ± 11.05	33.00 ± 2.61
10	linear_huber	Frozen	146.40 ± 15.93	34.00 ± 3.12
10	linear_huber	Scratch	144.08 ± 16.57	32.73 ± 6.79
10	linear_mae	Fine-tuned	137.55 ± 12.87	29.63 ± 2.73
10	linear_mae	Frozen	135.74 ± 14.12	30.82 ± 3.44
10	linear_mae	Scratch	135.23 ± 14.38	28.78 ± 3.05
10	log_fixed_huber	Fine-tuned	184.96 ± 9.99	38.53 ± 2.25
10	log_fixed_huber	Frozen	185.66 ± 14.61	39.28 ± 2.75
10	log_fixed_huber	Scratch	171.19 ± 14.04	35.00 ± 2.42
10	log_huber	Constant	181.05 ± 2.13	37.91 ± 0.54
10	log_huber	Fine-tuned	143.05 ± 12.49	29.40 ± 3.07
10	log_huber	Frozen	137.43 ± 13.49	29.24 ± 2.67
10	log_huber	Scratch	142.34 ± 11.06	28.12 ± 2.31

Table S8. Capacity and distinct fitting records; seed 42

Dataset	Budget	Policy	Total	Trainable	Encoder	Head	Records
co_ethylene	24	Fine-tuned	101002	101002	31144	69858	19
co_ethylene	24	Frozen	101002	69858	31144	69858	19
co_ethylene	24	Generic SSL	91714	91714	21856	69858	19
co_ethylene	24	Scratch	101002	101002	31144	69858	19
co_ethylene	5	Fine-tuned	101002	101002	31144	69858	4
co_ethylene	5	Frozen	101002	69858	31144	69858	4
co_ethylene	5	Generic SSL	91714	91714	21856	69858	4

Dataset	Budget	Policy	Total	Trainable	Encoder	Head	Records
co_ethylene	5	Scratch	101002	101002	31144	69858	4
nh3_h2	57	Fine-tuned	117514	117514	31272	86242	46
nh3_h2	57	Frozen	117514	86242	31272	86242	46
nh3_h2	57	Generic SSL	108098	108098	21856	86242	46
nh3_h2	57	Scratch	117514	117514	31272	86242	46
nh3_h2	5	Fine-tuned	117514	117514	31272	86242	4
nh3_h2	5	Frozen	117514	86242	31272	86242	4
nh3_h2	5	Generic SSL	108098	108098	21856	86242	4
nh3_h2	5	Scratch	117514	117514	31272	86242	4
sensor_drift	1	Fine-tuned	29542	29542	24992	4550	6
sensor_drift	1	Frozen	29542	4550	24992	4550	6
sensor_drift	1	Scratch	29542	29542	24992	4550	6
sensor_drift	20	Fine-tuned	29542	29542	24992	4550	96
sensor_drift	20	Frozen	29542	4550	24992	4550	96
sensor_drift	20	Scratch	29542	29542	24992	4550	96
sensor_drift	20	Fine-tuned	29542	29542	24992	4550	96
sensor_drift	20	Frozen	29542	4550	24992	4550	96
sensor_drift	20	Scratch	29542	29542	24992	4550	96

Architecture counts are constant for each dataset/policy. Fitting exposure counts can vary with selected repeated coordinates; complete per-seed counts are saved in statistics/model_size.csv. Parameter/exposure ratios are descriptive and do not alone quantify generalization.

NH3/H2 dataset and acquisition details

Sensor array

We use an array of 20 MEMS metal-oxide gas-sensor channels based on SnO2 nanowires with Pt/Ag surface modification. The sensors operate using a thermal-gradient sensing principle and provide complementary responses to NH3 and H2. The chamber accommodates the array in a 5 x 4 arrangement. We monitor the channels in four groups: CH1-CH5, CH6-CH10, CH11-CH15 and CH16-CH20, with electrical supply settings in the range 3.0-3.4 V.

Gas chamber and delivery system

We deliver NH3 and H2 to the sensor chamber through an automated mass-flow-controlled mixing system. The apparatus comprises gas supplies, background-gas delivery, mass-flow controllers, a switching valve, the sensor chamber, a power supply and a computer for control and recording. The switching valve directs the prepared mixture to the chamber during exposure. The outlet is connected to an exhaust line, and a cleaning-gas flow supports recovery between exposures. Figure S2 shows the system layout; Fig. S3 shows the sensor hardware, chamber assembly and mass-flow-controller installation.

Acquisition protocol

We programmed single-gas and binary NH3/H2 conditions spanning 0–500 ppm for each gas, with nonzero settings of 25, 50, 100, 200, 300, 400 and 500 ppm. Each sequence starts at gas delivery and records the complete 180-s exposure plus 700-s recovery cycle (880 s total). The acquisition interface was configured for a 2.0-s sampling interval. Acquisition took place over six days, but the six Combination-ID partitions were not intended as day splits. Day-by-day mixture details were not logged, preventing a reliable exposure-to-day mapping and leave-one-day-out evaluation. Temperature and relative humidity were neither controlled nor measured. These facts are clarified by the acquisition team; no timestamp-based cadence validation or retrospective environmental measurements are claimed.

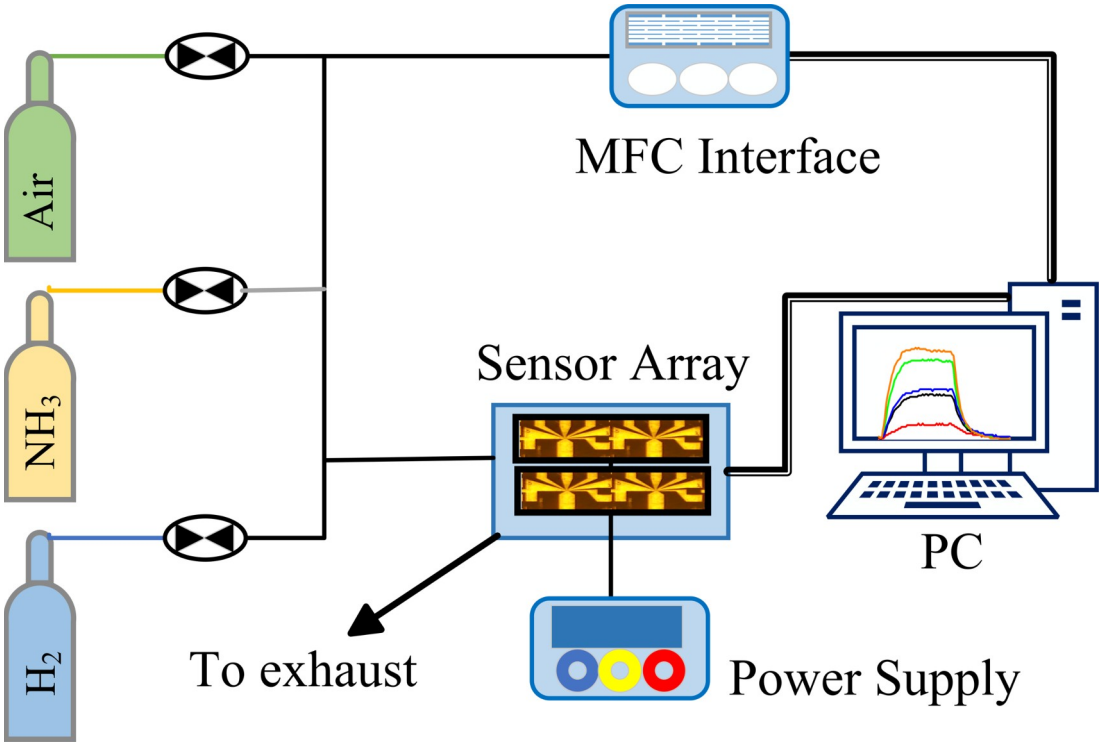


Fig. S2. Layout of our NH3/H2 mixing and measurement system, including gas supplies, mass-flow control, sensor array, power supply, exhaust and computer interface.



(a) MEMS sensor hardware (b) Sensor chamber (c) Mass-flow controllers

Fig. S3. Photographs of their experimental hardware: (a) sensor devices, (b) the chamber assembly containing the sensor array, and (c) the mass-flow-controller installation.

Revision 3.0: additional analysis and protocol clarification

Acquisition independence and environmental limitations

Identifier separation and stored-signal fingerprint checks establish digital separation only. Acquisition days may be shared across evaluation roles; without exposure-to-day records, overlap cannot be quantified. The existing folds support held-out identifier evaluation, not verified day independence. Environmental variation cannot be quantified retrospectively. Future

experiments will measure and record temperature and humidity; exposure-linked day/session logs are needed to enable day-held-out validation.

Physical timing and exact window definitions

The configured interval is 2.0 s. Aggregate counts give 162,404/378 = 429.64 observations per 880-s sequence, corresponding approximately to 2.048 s per observation. This is consistent with the configured interval, but is not a timestamp measurement. A 64-sample window represents about 128 s nominally or 131 s by this aggregate calculation (71–73% of the 180-s exposure); the first-to-last timestamp span at nominal cadence is 63 intervals, or 126 s. The first sample marks gas-delivery onset.

For an L-sample sequence, with zero-based indices: early windows start at offsets 0–4; middle windows start at floor((L−68)/2) plus offsets 0–4; late windows start at L−68 plus offsets 0–4. Each has 64 consecutive samples. Tests use only offset zero. Full-cycle windows linearly interpolate the complete interval [offset, L−1] to 64 points, using offsets 0–4 for pretraining/fitting and zero for testing. Full-cycle inputs include gas exposure and recovery. For approximately 430-sample cycles, middle and late windows predominantly probe recovery. Scalers are refitted using the corresponding pretraining windows; the primary early window remains predefined.

Multiplicity sensitivity

Table S9 retains equal fold weighting and independently resamples paired seed and fold axes. The 98.75% marginal percentile intervals implement a Bonferroni sensitivity for four comparisons using 200,000 draws (RNG seed 42). A second 200,000-draw run with RNG seed 2026 gives the same zero-inclusion decisions; endpoint shifts are at most 0.15 ppm. These are conditional descriptive bootstrap intervals. Bonferroni adjustment does not repair the limited number of chronological folds, overlapping training data, or unavailable acquisition-day labels. The original unadjusted Table I remains unchanged.

Table S9. Four-comparison multiplicity sensitivity; scratch minus fine-tuned MAE (ppm).

Gas	Difference	98.75% interval	Test folds
NH3	4.31	[-0.60, 9.46]	6
H2	10.58	[2.08, 21.54]	6
CO	24.74	[8.63, 41.71]	3
Ethylene	0.672	[0.167, 1.249]	3

NH3 includes zero after adjustment; H2, CO and ethylene remain positive. CO/ethylene intervals are indicative with only three test folds. NH3 is also the weakest case at five pairs and for sensor-ID/attention ablations.

CO ablation uncertainty

Table S10. Full-budget CO ablation-minus-full-model MAE: exploratory paired-seed percentile bootstrap, 200,000 draws, RNG seed 42, unadjusted 95% intervals. Seeds remain paired; test exposures are not resampled as independent acquisitions.

Removed	Mean ± SD (ppm)	95% interval (ppm)
Attention	10.27 ± 12.18	[2.68, 16.94]
Temporal mask	7.93 ± 14.86	[-0.33, 17.00]

The temporal-mask contrast includes zero. The attention contrast is positive in this conditional analysis, so it should not be called a statistical null solely because its mean is smaller than one SD. Both have substantial variability and neither supports a universal component benefit.

Representative pretraining compute

Representative seed-42 pretraining reruns on one NVIDIA GeForce RTX 3060 took 38.8 s for NH3/H2 (1,260 windows) and 19.9 s for CO/ethylene (735 windows), each for 60 epochs with batch size 64. Timings include model initialization and optimization but exclude data loading, scaling and checkpoint writes. They are per-model measurements, not reconstructed total campaign costs; checkpoints are reused across adaptation budgets. Software and timing details are supplied in the supplement.

Timing environment: Python 3.13.2, PyTorch 2.12.1+cu126, CUDA 12.6. One GPU was used. The reported reruns use the original pretraining implementation and fixed-split preprocessing, without adaptation training or reuse of a cached pretrained state. The combined measured time for these two representative runs is 58.6 s; it is not the total ten-seed/multi-fold experiment cost. Descriptor-benchmark pretraining uses 40 epochs (batch size 256); its wall time was not measured in this raw-response timing check. Hardware/software and source hashes are saved in evidence/compute_measurement.json.

Loss denominator guard

Equation (14) guards both denominators. The V3.0 raw-response implementation adds clamp_min(1) to the visible denominator. With at most three spans of length 16 in a 64-sample window and at most three entirely hidden channels, at least 16 temporal positions remain visible in each nonhidden channel. The visible denominator is therefore always positive for the reported 16-/20-channel protocols. This defensive code change leaves their loss unchanged; historical predictions and results are retained.